Module 2 – Frontend – HTML

HTML Basics

• Question 1: Define HTML. What is the purpose of HTML in web development?

HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages. It uses tags and elements to define the content of a webpage such as headings, paragraphs, images, links, and more.

The main purpose of HTML in web development is to **structure the content of a website**. It acts as the backbone of any webpage by organizing text, images, and other elements so that browsers can display them properly. HTML works together with CSS (for styling) and JavaScript (for functionality) to create complete and interactive websites.

• Question 2: Explain the basic structure of an HTML document. Identify the mandatory tagsand their purposes.

An HTML document has a standard structure that every webpage follows. This structure helps the browser understand how to display the content.

The basic structure includes:

- <!DOCTYPE html>
  This declaration defines the document type and tells the browser that the document is HTML5.
- <html>
  This is the root element that contains the entire HTML document.
- <head>
  This section contains meta-information about the webpage such as title, character set, and links to stylesheets.
- <title>
  It defines the title of the webpage, which appears on the browser tab.
- <body>
  This is the main part of the document where all visible content like text, images, and links are placed.

These tags are considered mandatory because without them, the browser may not correctly interpret or display the webpage.

• Question 3: What is the difference between block-level elements and inline elements in HTML? Provide examples of each.

HTML elements are categorized into block-level and inline elements based on how they display on a webpage.

Block-level elements:

- They take up the full width of the page
- Start on a new line
- Used to structure the layout of the page

Examples include <div>, <p>, <h1> to <h6>

Inline elements:

- They only take up as much width as needed
- Do not start on a new line
- Used within block elements to style small parts of content

Examples include <span>, <a>, <strong>, <em>

In simple terms, block elements create structure, while inline elements format content within that structure.

• Question 4: Discuss the role of semantic HTML. Why is it important for accessibility andSEO? Provide examples of semantic elements.

Semantic HTML refers to the use of HTML elements that clearly describe their meaning and purpose in the webpage. Instead of using generic tags like <div>, semantic elements explain the role of the content.

Examples of semantic elements include:

- <header> for the top section
- <nav> for navigation links
- <section> for grouping content
- <article> for independent content
- <footer> for the bottom section

Importance for Accessibility:
Semantic HTML helps screen readers understand the structure of the webpage, making it easier for visually impaired users to navigate and access content.

Importance for SEO:
Search engines like Google can better understand the content of the page when

semantic elements are used. This improves the ranking of the website in search results.

In simple terms, semantic HTML improves both **user experience and search engine visibility**, making websites more accessible and optimized.

HTML Forms

• Question 1: What are HTML forms used for? Describe the purpose of the input, textarea, select, and button elements.

HTML forms are used to **collect user data** and send it to a server for processing. They are commonly used in login pages, registration forms, search bars, feedback forms, and more.

Different elements are used inside forms to take various types of input:

- **<input> element**
  This is the most commonly used form element. It is used to take user input in different formats such as text, password, email, number, checkbox, radio button, etc. The type attribute defines the kind of input.
- **<textarea> element**
  It is used for multi-line text input. It is suitable for longer content like comments, feedback, or messages.
- **<select> element**
  It is used to create a dropdown list. Users can select one or more options from a list defined using <option> tags.
- **<button> element**
  It is used to perform actions like submitting the form, resetting data, or triggering scripts. The most common type is the submit button, which sends form data to the server.

These elements together make forms interactive and useful for collecting different types of user information.

• Question 2: Explain the difference between the GET and POST methods in form submission. When should each be used?

GET and POST are two methods used to send form data from the client to the server.

GET method:

- Sends data through the URL
- Data is visible in the browser address bar
- Limited data size
- Can be bookmarked
- Less secure

POST method:

- Sends data in the request body
- Data is not visible in the URL
- Can handle large amounts of data
- More secure than GET
- Cannot be bookmarked

When to use:

- Use **GET** when requesting data without changing anything (like search queries)
- Use **POST** when sending sensitive data (like passwords) or when data needs to be stored or processed

• Question 3: What is the purpose of the label element in a form, and how does it improve accessibility?

The **<label> element** is used to define a text label for a form input field. It describes what the input field is for, such as “Username” or “Email”.

The label is connected to an input field using the **for attribute**, which matches the id of the input element.

Importance:

- It makes forms easier to understand for users
- Improves usability by allowing users to click on the label to focus the input field
- Enhances accessibility for screen readers, helping visually impaired users understand the form

In simple terms, the label element improves both **user experience and accessibility** by clearly describing form fields.

HTML Tables

• Question 1: Explain the structure of an HTML table and the purpose of each of the following elements:<table>,<tr>,<th>,<td> and <thead>.

An HTML table is used to **organize data in rows and columns**. It is commonly used for displaying structured data like schedules, reports, or lists.

The main elements of a table are:

- **<table>**
  This is the main container that defines the table. All table content is placed inside this tag.
- **<tr> (Table Row)**
  It is used to create a row in the table. Each row can contain multiple cells.
- **<th> (Table Header)**
  It defines a header cell in a table. The content inside <th> is usually bold and centered. It is used for column headings.
- **<td> (Table Data)**
  It defines a standard data cell in a table. It contains the actual data of the table.
- **<thead>**
  It is used to group the header section of the table. It usually contains <tr> and <th> elements. It helps in better structure and styling of the table.

Together, these elements help in creating a well-structured table for displaying data clearly.

• Question 2: What is the difference between colspan and rowspan in tables? Provide examples.

Both **colspan** and **rowspan** are attributes used in table cells to merge cells.

- **colspan**
  It is used to merge two or more columns into a single cell.
  Example: If a cell spans across 2 columns, we use colspan="2".
- **rowspan**
  It is used to merge two or more rows into a single cell.
  Example: If a cell spans across 2 rows, we use rowspan="2".

Difference:
Colspan works horizontally (across columns), while rowspan works vertically (across rows).

• Question 3: Why should tables be used sparingly for layout purposes? What is a better alternative?

Tables should be used sparingly for layout because they are meant for **displaying data**, not for designing page layouts. Using tables for layout makes the code complex, harder to maintain, and less flexible. It also affects website performance and accessibility.

A better alternative is using **CSS layout techniques** such as:

- Flexbox
- Grid system

These methods are more flexible, responsive, and easier to manage. They help create modern and responsive web designs without misusing tables.